

Server setup

VPS Setup & Application Hosting Guide

0. Prerequisites

Ensure the following before starting:

- VPS with Ubuntu 20.04 or later
 - Domain pointing to VPS IP
 - Access to Git repository
 - SSH client (Terminal or PuTTY)
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1. Connect to VPS

```
1 | ssh root@your_server_ip
```

2. Initial Server Setup

Update system

```
1 | apt update && apt upgrade -y
```

Create a non-root user (recommended)

```
1 | adduser deployer
2 | usermod -aG sudo deployer
```

Enable firewall

```
1 | ufw allow OpenSSH
2 | ufw allow 'Nginx Full'
3 | ufw enable
```

3. Install Required Software

Install Node.js (LTS)

```
1 | curl -fsSL https://deb.nodesource.com/setup_18.x | bash -
2 | apt install -y nodejs
```

Install Git

```
1 | apt install git -y
```

Install PM2

```
1 npm install -g pm2
```

Install Nginx

```
1 apt install nginx -y
```

4. Setup SSH for GitHub (Private Repository)

```
1 ssh-keygen -t ed25519 -C "your-email@example.com"  
2 cat ~/.ssh/id_ed25519.pub
```

Add the generated public key to GitHub under SSH Keys.

5. Clone Project Repository

```
1 git clone git@github.com:your-username/your-repo.git  
2 cd your-repo
```

6. Install Project Dependencies

```
1 npm install
```

7. Environment Variables

Create a `.env` file:

```
1 nano .env
```

Example:

```
1 MONGODB_URI=your_db_uri  
2 SUPER_ADMIN_EMAIL=your_email  
3 SUPER_ADMIN_PASSWORD=strong_password  
4  
5 RAZORPAY_KEY_ID=your_key  
6 RAZORPAY_KEY_SECRET=your_secret  
7  
8 AWS_SES_ACCESS_KEY=your_key  
9 AWS_SES_SECRET_KEY=your_secret  
10 AWS_SES_REGION=ap-south-1  
11 AWS_SES_FROM_EMAIL=no-reply@yourdomain.com
```

Important guidelines:

- Do not commit `.env` files to Git
 - Do not share credentials in documentation
 - Use separate credentials for development and production
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8. Build Application (Production)

```
1 npm run build
```

9. Run Application with PM2

```
1 pm2 start npm --name "nextjs-app" -- start
```

Save process:

```
1 pm2 save
2 pm2 startup
```

10. Configure Nginx (Reverse Proxy)

Create configuration file:

```
1 nano /etc/nginx/sites-available/nextjs
```

Add the following configuration:

```
1 server {
2     listen 80;
3     server_name yourdomain.com;
4
5     location / {
6         proxy_pass http://localhost:3000;
7         proxy_http_version 1.1;
8         proxy_set_header Upgrade $http_upgrade;
9         proxy_set_header Connection 'upgrade';
10        proxy_set_header Host $host;
11        proxy_cache_bypass $http_upgrade;
12    }
13 }
```

Enable configuration:

```
1 ln -s /etc/nginx/sites-available/nextjs /etc/nginx/sites-enabled/
2 nginx -t
3 systemctl restart nginx
```

11. Enable HTTPS (SSL)

Install Certbot:

```
1 apt install certbot python3-certbot-nginx -y
```

Run SSL setup:

```
1 certbot --nginx -d yourdomain.com
```

Verify auto-renewal:

```
1 | certbot renew --dry-run
```

12. Deployment Workflow

For updating application:

```
1 | git pull
2 | npm install
3 | npm run build
4 | pm2 restart nextjs-app
```

13. Monitoring and Logs

```
1 | pm2 status
2 | pm2 logs
3 | pm2 monit
```

14. Best Practices

- Use a non-root user for deployments
 - Always run application in production mode
 - Secure environment variables
 - Configure firewall and HTTPS
 - Regularly monitor logs and application status
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15. Common Issues

- Application running in development mode instead of production
 - Incorrect environment variables
 - Nginx misconfiguration
 - Port conflicts (default: 3000)
 - SSL not properly configured
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16. Optional Enhancements

For automation and CI/CD, consider using:

- GitHub Actions
 - Jenkins
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